

Hemant Pathak

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Education

Indian Institute of Technology (ISM) Dhanbad

Integrated M.Tech in Mathematics and Computing, GPA: 8.80 / 10.00 2023 – 2028

Relevant Coursework

Database Management Systems • Statistical Inference • Object-Oriented Programming • Operating Systems
• Theory of Computation • GPU Computing • Linear Algebra

Technical Skills

Languages:	Python (async, typed), JavaScript, C++, Java, C, MySQL, PostgreSQL
AI/ML Stack:	TensorFlow, PyTorch, Scikit-Learn, LangChain, LangGraph, OpenAI API, Azure OpenAI, OpenCV, NumPy, Pandas, FastAPI, FAISS
Core Concepts:	Prompt Engineering & Development, NLP Pipelines, RAG (Retrieval-Augmented Generation), One-Shot & Multi-Shot Inference, Autonomous Agent Frameworks, LLM Orchestration, Embeddings, Computer Vision, OCR, Model Evaluation & Performance Tuning
Tools/Cloud:	Microsoft Azure, Azure Doc Intelligence, AWS SageMaker, Docker, GitHub Actions CI, Git, Google Cloud

Experience

Apparent Energy Solutions Pvt. Ltd. Remote
AI Engineering Intern — Agentic MCP & Retrieval Systems Apr 2026 – Present

- Designed and shipped a **FastMCP server exposing 16+ autonomous tool-based APIs** for Claude Desktop, enabling multi-step LLM decision-making workflows and cutting token overhead by ~40% via one-shot query bundling (`get_meter_summary`).
- Engineered structured **prompt development strategies** (one-shot, multi-shot, fallback chains) to enforce reliable agent behavior, entirely eliminating SQL hallucinations via deterministic tool-invocation paths.
- Built a schema-aware retrieval pipeline with hierarchical tool orchestration (`describe_table` → `query_database`), resolving cascading timeouts via 30-second hard kills to stabilize transport.
- Validated system performance using a 48-case parametric test suite and multi-step prompt evaluations; tracked failure modes to optimize tool-call budgets and improve overall agent resilience.

Apparent Energy Solutions Pvt. Ltd. Remote
Lead Software Engineer Intern — AI & Computer Vision Jan 2026 – Mar 2026

- Orchestrated high-throughput Computer Vision pipelines on Google Street View and multi-spectral satellite imagery to geospatially map India’s electrical grid infrastructure.
- Trained and optimized a custom **YOLOv11** model (TensorRT inference) for pole detection with hybrid instance segmentation to delineate overhead wire trajectories.
- Applied graph-theoretic algorithms with **GeoPandas** and **PostGIS** to model pole-to-pole connectivity, establishing a digital twin of the grid topology.

Aeroqube Inc. Remote
Software Engineering Intern (AI) Aug 2025 – Dec 2025

- Built multimodal autonomous reasoning agents using **LangChain**, **LangGraph**, and **Med Gemma 27B** with a FAISS-backed i-MedRAG retrieval pipeline (RAG + embeddings); engineered **NLP-driven prompts** with one-shot and multi-shot patterns to synthesize context-aware clinical summaries.
- Designed end-to-end data preprocessing and feature engineering pipelines for volumetric CT/X-ray datasets; fine-tuned **MED-SAM 2** for zero-shot lesion segmentation, achieving >90% Dice similarity coefficient (DSC) on held-out evaluation sets.
- Containerized and deployed high-throughput inference services on **AWS SageMaker** with Docker and model

quantization for low-latency production serving; monitored model performance metrics post-deployment to ensure reliability.

Technod8.ai

Remote

Generative AI Intern

Jul 2025

- Fine-tuned **YOLOv10** on custom-annotated datasets for Document Layout Analysis on multilingual corpora (NCIIPC AI Grand Challenge); performed data cleaning, class-balanced augmentation, and feature engineering to maximize mAP, integrating with **PaddleOCR** for downstream **NLP**-based semantic parsing and information extraction.

Achievements

- Top **1% finish (Rank 80)** among 8,000+ participants — **Shell.ai Blend Fuel Price Prediction Challenge 2025**; developed and tuned a production-grade ML model with rigorous data preprocessing, feature selection, and algorithm benchmarking using **Scikit-Learn** and **TensorFlow**.
- **Runner-up**, Algoverse algorithmic trading hackathon; statistical arbitrage strategy outperformed 50+ teams.

Positions of Responsibility

- **Student Placement Coordinator**, Career Development Centre, IIT (ISM) Dhanbad
- **Member**, Fintech Club, IIT (ISM) Dhanbad